DATA BASES – PROF. D. M. BRAGA – A. A. 2021/22

CALL 4 – JULY 19TH 2022 – DURATION: 1h 30m

SQL (24 points)



The schema below describes the articles of several online newspapers and the comments to the articles, as posted by their readers. A reporter may collaborate with more than one newspaper. Articles are identified by the newspaper, the date, and progressive number for that edition of the newspaper.

REPORTER (ReporterId, Surname, FirstName, BirthDate, Nationality)
NEWSPAPER (NewspaperName, Director, City, Publisher)
ARTICLE (NewspaperName, Date, ArticleNumber, Title, ReporterId, NumberOfWords)
COMMENT (CommentId, NewspaperName, Date, ArticleNumber, UserNickname, Text)

1. Briefly but precisely explain what is extracted by the following view [**5 p**]:

a.
```
create view acc( Nwsp, Rep, Ppcnt, TotWds ) as
    select NewspaperName, ReporterId,
           count(*), sum(NumberOfWords)
    from ARTICLE natural join REPORTER
    group by NewspaperName, ReporterId
```

b.
```
select Date, ReporterId, count( * )
from ARTICLE N natural join COMMENT C
group by ReporterId, Date
having count( * ) >= ALL ( select count( * )
                           from ARTICLE natural join COMMENT
                           where Date = N.Date
                           group by ReporterId )
```

2. Extract in SQL:

- **a**. The average "verbosity" of the journalists of each nationality (like: *Italians* 600 words per paper, *Germans* 820 w/p, *Belgians* 395 w/p, …); [**5 p**]
- **b**. The *youngest* Non-Italian reporter to have an article published on "PoliNews", considering the age of the author at publication time (*assume for simplicity that a subtraction between two dates returns an interval expressed as the number of days in between*); [**5 p**]
- **c**. The reporters who wrote for at least 3 Newspapers and never reached the goal of 100 comments to one single article. [**5 p**]

3. Assume that the view defined in **1.a** is materialized. Formulate a set of triggers that incrementally maintain the materialization of the view in response to *insertions* on table ARTICLE (all other events can be neglected) [**4 p**]

Conceptual Design (8 points)

4. Design an E-R model for the database described in the following scenario:

A travel agency organizes trips for both business and leisure travellers. Each customer chooses a destination, specifies the number of people who will travel with them, chooses the transportation mode (train, plane, ship) and a preferred category of the hotel. Depending on the destination, the agency also takes care of issuing the entry visas (by requesting the visas of all the participants to the trip from the competent embassy); to do so, the agency collects the travellers' passports - containing their names, serial numbers, and dates of issue and expiration - and then returns the visas to the customers.

Depending on the availability, journeys can be without stops or may include some intermediate stops: in this case, the arrival and departure times of each stop are known. The hotels at which the reservations will be made may be of the requested category, or of a higher one, and are characterized by a name, an address, and their average price per night (but for each specific booked stay the actual applied price is known).

- *The project is evaluated for completeness, correctness, legibility, minimality and self-explanatory nature.*
- *Remember to specify the **identifiers** of all entities and the **cardinalities** of all associations.*